NyaySaathi – AI-Powered Legal Assistance Platform for Consumer Disputes in India

1. Introduction and Objective

NyaySaathi is an AI-powered legal assistance platform designed to help ordinary citizens in India understand and address consumer-related legal problems. People frequently face issues such as defective products, delayed refunds and disputes with online sellers but lack legal awareness or cannot afford immediate legal consultation. The platform provides accessible legal information through a conversational interface, explains relevant laws in simple language and assists users in organising case information and preparing documents.

The current implementation focuses on consumer disputes and refund complaints. Other areas, including criminal, family, property, employment and taxation matters, are outside its scope and require professional legal assistance.

2. System Architecture and Working

The application follows a structured processing pipeline:

1. **User Input:** Accepts queries in English, Hindi and Hinglish through text or voice input.
2. **Safety and Triage:** Identifies the nature of the dispute, checks whether it falls within the supported domain and asks questions when important information is missing.
3. **Legal Retrieval:** Uses BM25-based keyword search to retrieve relevant provisions from a knowledge base containing 37 provisions from 9 official documents.
4. **AI Response Generation:** Uses an AI model to understand the user's problem, extract facts and explain relevant legal provisions in simple language.
5. **Automated Validation:** Checks generated responses for unsupported legal claims. If validation fails, one correction attempt is made; otherwise, the response is blocked.
6. **Final Output:** Displays the answer, supporting legal sources and a confidence indicator representing the quality of supporting information, not the probability of winning a case.

3. Major Features

- **Legal Chat Assistant:** Provides interactive assistance for consumer disputes in English, Hindi and Hinglish.
- **Build My Case:** Creates an editable case file separating user-provided facts, AI inferences and unverified information.
- **Know My Rights:** Displays official legal provisions, simplified explanations, source links and suggested actions.
- **Evidence Checklist:** Helps users identify and organise documents required to support complaints.
- **Generate Documents:** Creates five types of preliminary legal documents in TXT or DOCX format. All outputs are marked as drafts and require user review.
- **Practice My Argument:** Provides simulated conversations and feedback to help users present their disputes effectively.

4. Technology Stack and Implementation

The application is developed using **Python and Streamlit**, with Google Gemini API for natural language processing and response generation. BM25 is used for keyword-based legal retrieval, while structured data organises legal provisions. The application also supports Gemini, OpenAI and other OpenAI-compatible APIs through environment-based configuration.

Development involved implementing application modules, API integration, legal-source retrieval, response validation, document generation and interface design. Additional code was developed for automated testing, API error handling, retry mechanisms, resumable test execution and API key security.

Official legal documents were prioritised while preparing the knowledge base. A verification script checks legal quotations against their original sources. The system also accounts for differences between older legal summaries and updated provisions, including the ₹50 lakh jurisdictional threshold for District Commissions under the 2021 rules and the commencement date of a 2026 e-commerce amendment (1 January 2027).

5. Testing and Evaluation

The platform was evaluated through automated code testing, legal-source verification and AI response evaluation.

Evaluation Parameter	Result
Automated code tests	175/175 passed
Legal quotations verified through scripts	65/65 matched

Additional manual quotation check	1
AI test cases executed	8 out of 12
AI test cases passed / failed	3 / 5
Applicable AI evaluation checks passed	56/61
Citations delivered to users	38
Responses corrected / blocked	3 / 1

Four AI test cases were not executed due to free-tier API usage limitations. Testing covered defective products, refund complaints, multilingual interaction and out-of-scope queries. Formal legal review by a qualified lawyer has not yet been completed. Therefore, these results demonstrate technical functionality rather than guaranteed legal correctness.

6. Challenges and Limitations

Major development challenges included API server errors, usage restrictions, API key security, conflicting legal information and Streamlit interface compatibility. These were addressed through alternative model configurations, retry mechanisms, improved security checks, verification against primary legal documents and browser-based interface testing.

The current system has several limitations:

- Legal coverage is restricted mainly to consumer disputes.
- Keyword-based retrieval may miss relevant provisions when users use different terminology.
- Automated validation cannot guarantee correct legal interpretation.
- RBI refund regulations and the RBI Ombudsman framework are not included.
- Legal amendments require periodic verification.
- Voice functionality has not been tested using a real microphone.
- Some tests revealed premature responses and overly restrictive refusals.
- Generated legal advice has not undergone formal lawyer review.

7. Scalability and Future Scope

The application can be expanded by adding verified legal documents, keywords, scope rules and test cases. Its AI model can also be changed through configuration without major modifications to the application code.

Future deployment improvements include API-based backend architecture, multiple application instances, caching, rate limiting, request queues and semantic search. Further development will focus on improving fact collection before providing guidance, correcting validation issues, completing pending tests, expanding legal coverage, obtaining professional legal reviews and conducting real-user testing.

During testing, eight evaluation cases used 49 AI calls, approximately 102,591 input tokens and 22,735 output tokens. Based on the pricing considered during development, the estimated API cost was US$0.02 per case (approximately US$185 for 10,000 cases), potentially increasing to US$0.03 per case (US$340 for 10,000 cases) from January 2027. These estimates exclude hosting expenses.

8. Conclusion

NyaySaathi demonstrates the integration of conversational AI, structured legal information retrieval and automated validation to improve access to basic legal assistance. Its features help users understand consumer rights, organise case information, prepare documents and practise legal arguments. Although the initial implementation demonstrates functional application modules and successful automated code testing, further improvements in legal accuracy, knowledge-base coverage and professional verification are necessary before large-scale deployment.